

Sujatro Ganguli

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PROFESSIONAL SUMMARY

Third Year Computer Science student proficient in Python, AI Engineering, and Web Development. Expert in RAG frameworks and open-source libraries. Proven leader, as Technical Head at SEDS VIT, delivering high-availability platforms and automated research software.

EDUCATION

Vellore Institute of Technology (VIT), Vellore — B.Tech in CSE (CGPA: 8.59) **2023 – Present**

TECHNICAL SKILLS

Languages: Python (Expert), C++, C, Java, JavaScript, TypeScript, SQL, Quickshell

Frameworks & Tools: React, Next.js, TanStack, Django, FastAPI, Docker, CI/CD, Git, OpenCV, Wayland, Qt/QML, LangGraph, SQLite, GitHub Actions, AWS (EC2, Lambda, RDS)

Domains: RAG Systems, LangGraph Agents, Vector Indexing, Full-Stack Dev, ML Optimization, Linux Systems

EXPERIENCE

- **Web Development Intern — Terionix:** Architected a full-stack platform (Django, TanStack) including an admin portal and custom web scrapers. Containerized the architecture via Docker and established a CI/CD pipeline deploying to a custom mobile-based staging server for end-to-end testing. *May 2026 – Present*
 - **Technical Head — SEDS VIT:** Architected event portals for Hermes 🌐 and Erebus 🌐; managed **270+ registrations** with **100% uptime** via Vercel and Cloudflare DNS. *2026*
 - **Research Intern — PRL:** Automated SHARAD radar visualization using Python; reduced rendering latency for **3 GB of data** from hours to **sub-10 seconds**. *2025*
 - **Web Volunteer — Outreach:** Redesigned websites for two NGOs using HTML/CSS; improved accessibility and dedicated **100+ hours** to community initiatives with NGO CINI. *2021 – 2023*
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TECHNICAL PROJECTS

- **Vessel:** Local-first, configurable knowledge workspace featuring an offline LangGraph-powered RAG pipeline for on-device querying of study materials and notes. *Tech: Python, LangGraph, Qt/QML, SQLite, RAG* 🌐
- **HyprQuickPaper:** Wayland wallpaper manager optimized for sub-200ms switching and 0% idle CPU usage. *Tech: Quickshell, Wayland, Shell, Linux* 🌐
- **EVA — RAG Tool:** Engineered .eva format for versioned indexing, reducing re-indexing time from hours to seconds. *Tech: Python, RAG Architecture, Vector Indexing, LLMs* 🌐
- **Gestualis:** Low-power gesture library using dot-product logic for 100% consistency. Automated package publishing via GitHub Actions, reaching 9,000+ downloads. *Tech: Python, OpenCV, Computer Vision, MediaPipe, GitHub Actions* 🌐
- **Automated Question Gen:** RAG pipeline automating academic assessment creation, reducing manual effort by 75%. *Tech: Python, RAG, NLP, AI Engineering* 🌐
- **Exoskeleton Optimization:** Applied NSGA-II via MATLAB to optimize biomechanical parameters in real-time. *Tech: MATLAB, NSGA-II, Optimization Algorithms* 🌐
- **Hermes Portal:** Architected event portal for SEDS VIT managing 270+ registrations with 100% uptime. *Tech: React, Next.js, Vercel, Cloudflare DNS* 🌐
- **Erebus Portal:** High-availability registration system for SEDS VIT events with integrated DNS management. *Tech: React, Next.js, Vercel, Cloudflare DNS* 🌐

- **OriginShift:** Algorithm designed to generate randomized yet guaranteed solvable mazes. *Tech: Python, Algorithms, Maze Generation, Data Structures* 🌐
 - **HackBattle (Tracker):** Hackathon project that tracks hand and eye movement to control mouse pointer navigation. *Tech: JavaScript, Computer Vision, Webcam API, Tracking* 🌐
 - **HyprTodo:** Todo management system utilizing shell scripts for task execution and Quickshell for the interface. *Tech: Shell Scripting, Quickshell, Wayland, Linux* 🌐
 - **Sentimental Text:** Modular CLI tool for sentiment analysis featuring preprocessing, feature extraction, and deep learning models. *Tech: Python, Deep Learning, NLP, CLI* 🌐
 - **Arabella:** AI-powered search tool utilizing a Graph Database to store and retrieve semantic connections. *Tech: Python, KuzuDB, GraphDB, Search Engine, AI* 🌐
 - **DSW (Room Booking):** Management system for handling room bookings and scheduling. *Tech: Web Development, Database Management, HTML/CSS* 🌐
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AWARDS & ACHIEVEMENTS

- **1st Place — Caterpillar Hackathon (2024):** Built technician app with face-ID and voice-to-text.
 - **Track Prize — Yantra Central Hackathon (2025):** Developed genomic analysis platform with **89% accuracy**.
 - **1st Place — Prima Vera Filmmaking (2023) — National Qualifier — First Lego League (2019)**
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CERTIFICATIONS & INTERESTS

Certifications: DELF B1 (French), Udemy Python Programming (64.5 hours)

Languages: English (Professional), French (Conversational), Hindi, Bengali

Interests: Linux System Customization, Machine Learning Research, Theatre